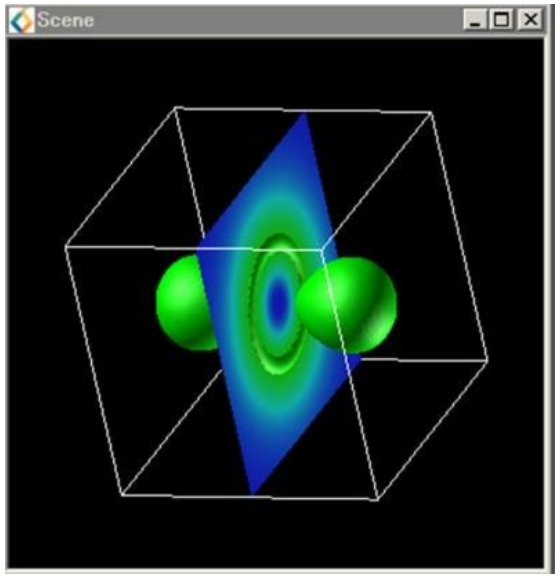
Volume Data

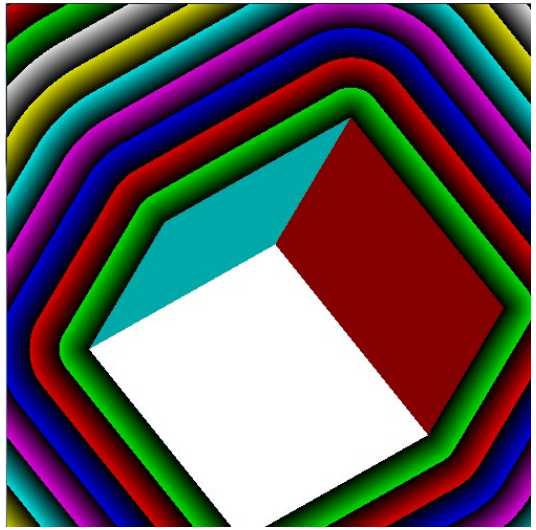
- 3D array of data. Each volume element (voxel) is a value representing some measurement or calculation
- e.g. Magnetic resonance imaging (MRI) data: Strong magnetic field aligns magnetization of hydrogen atoms, and then measures radio waves they emit (body tissue contains a lot of hydrogen (water)). Computed Tomography (CT) data: Interior images of the body are calculated using many X-rays of the body.

Volume Data

- e.g. simulation: probability of electron position around a H_2 molecule
- Machine representation simple: 3D array of char/integer/real, etc.
- Conversion to volume data: scanned (MRI/CT), numerical simulation (hydrogen data), or distance fields
- Rendering to screen (e.g. MIP – later slides)



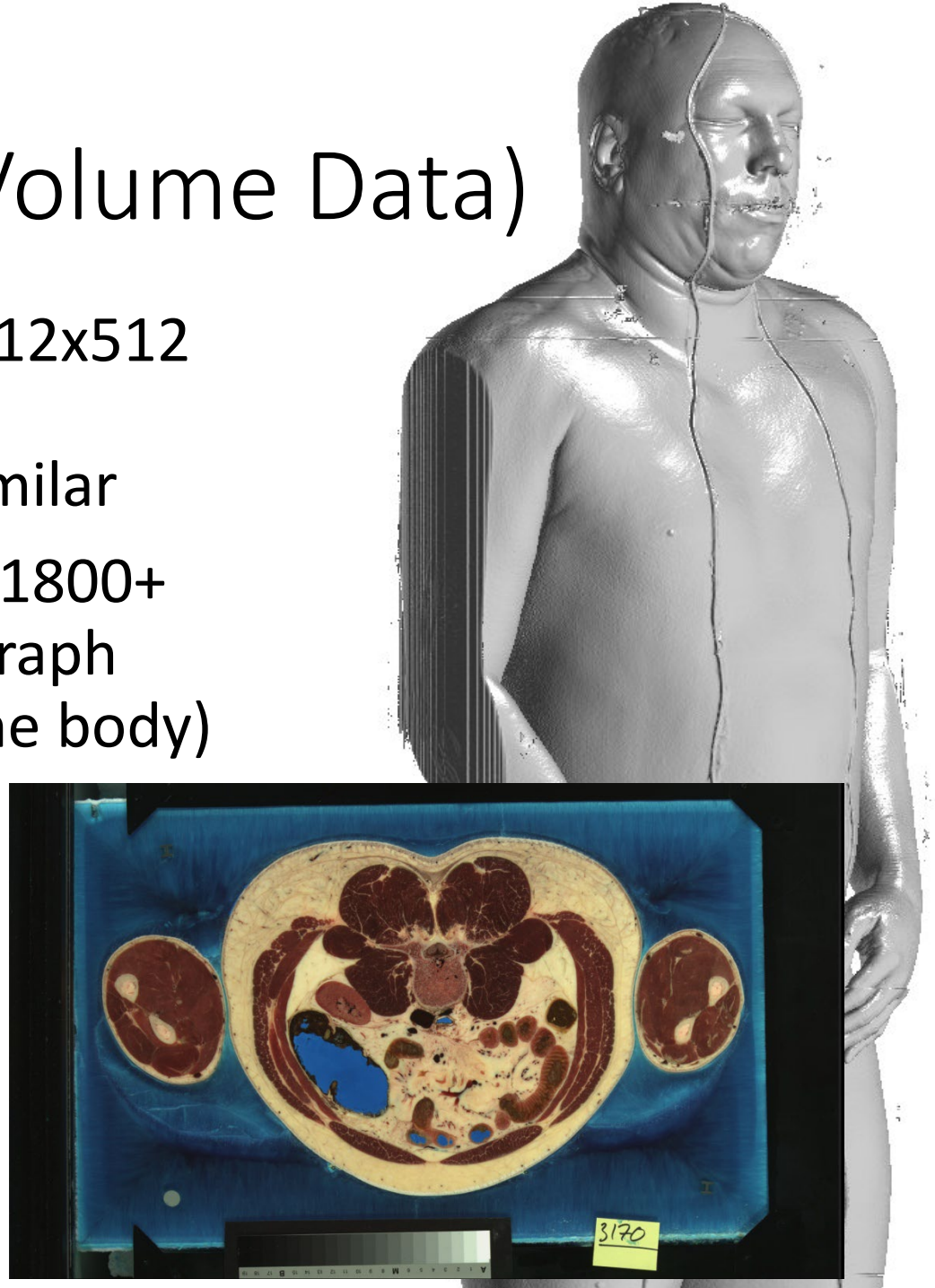
Hydrogen data



Distance around object

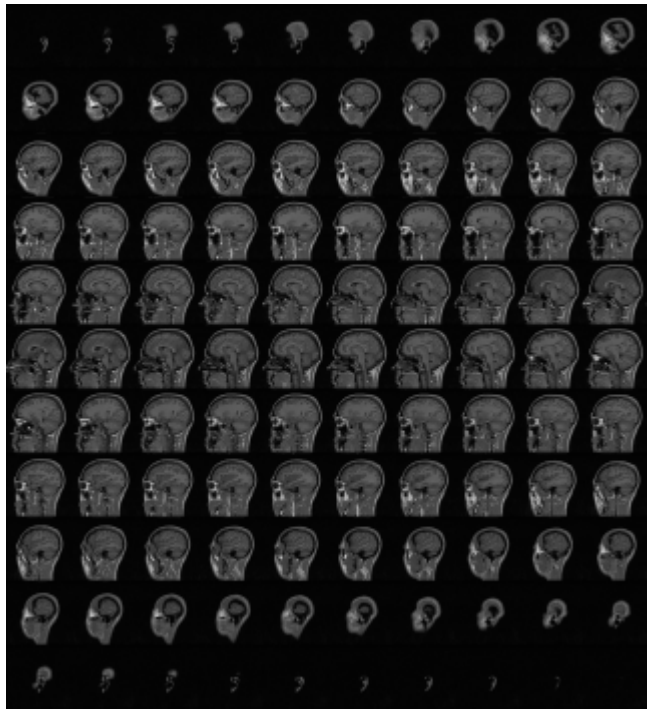
Visible Human Project (Volume Data)

- CT scan 1800+ slices, each 512x512 voxels. Each voxel is 2 bytes (integer) ~ 1GB. MRI scan similar
- Body then photographed at 1800+ 1mm intervals (each photograph represents a slice through the body)



Volume Rendering Input

- The data consists of a number of images representing slices through the body
- Each pixel in each image has the measured quantity (X-ray absorption for CT scans)
- Stacking the images creates a 3D array of “voxels” = volume elements (pixels=picture elements)

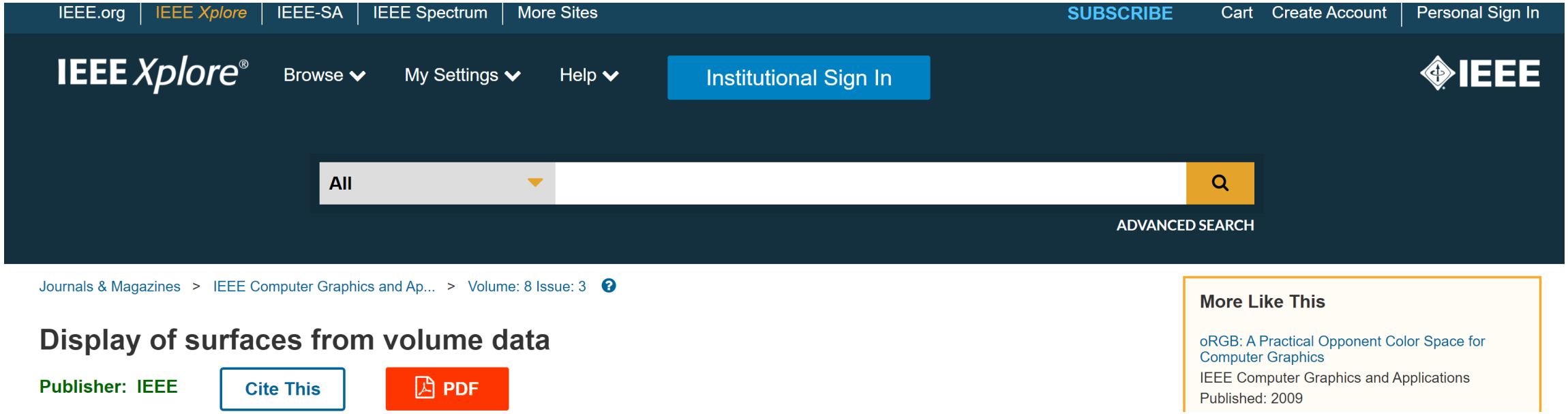


Volume Rendering (history)

- I wrote a chapter on Volume Ray Casting for my PhD thesis (Chapter published in a BCS conference). The earliest volume rendering papers are:
- M. Levoy. Display of surfaces from volume data. IEEE Computer Graphics and Applications, 8(3):29–37, May 1988.
- P. Sabella. A rendering algorithm for visualizing 3D scalar fields. In Proc. SIGGRAPH ’88 (Atlanta, Georgia, August 1-5, 1988), volume 22(4), pages 51–57. ACM SIGGRAPH, New York, August 1988.
- C. Upson and M. Keeler. V-Buffer : Visible volume rendering. In Proc. SIGGRAPH ’88 (Atlanta, Georgia, August 1-5, 1988), volume 22(4), pages 59–64. ACM SIGGRAPH, New York, August 1988.
- L. Carpenter R. A. Drebin and P. Hanrahan. Volume rendering. In Proc. SIGGRAPH ’88 (Atlanta, Georgia, August 1-5, 1988), volume 22(4), pages 65–74. ACM SIGGRAPH, New York, August 1988.

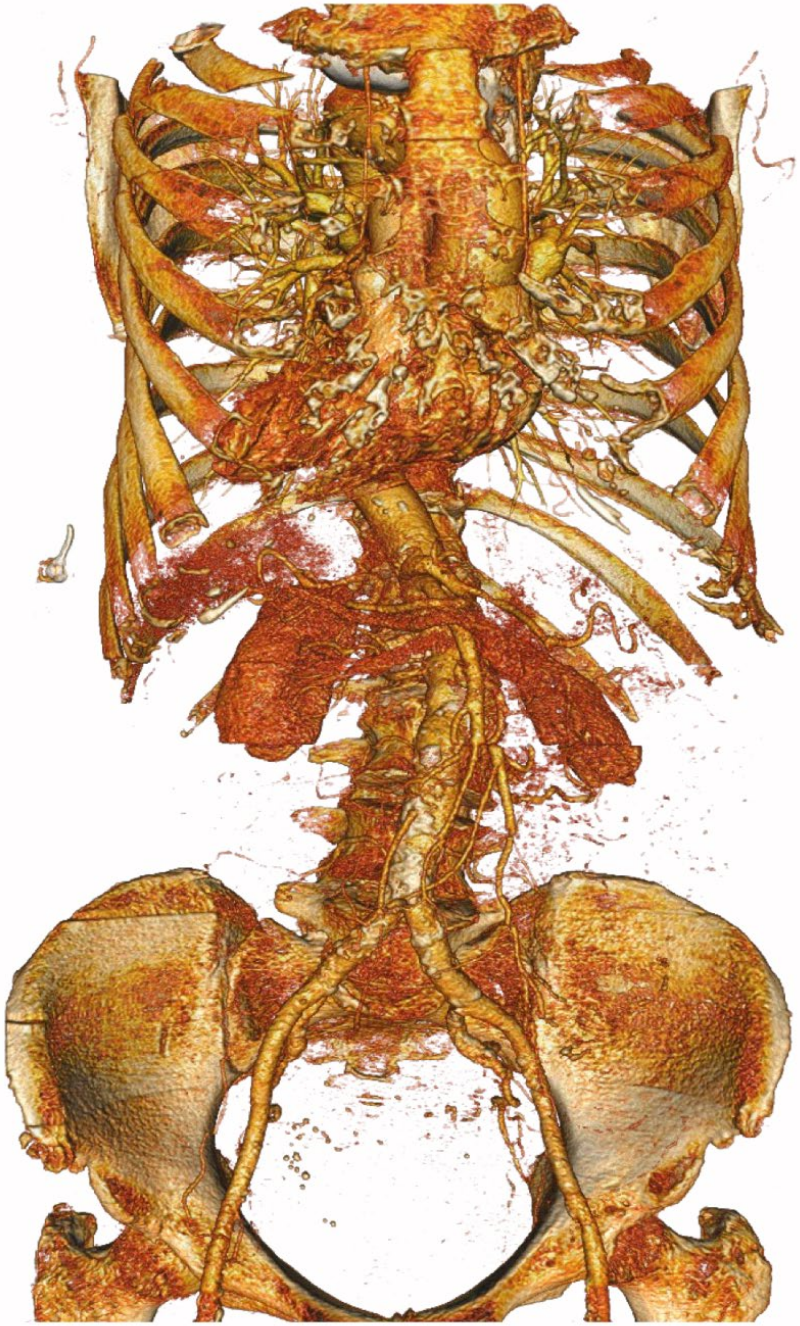
Levoy's Volume Rendering paper

- You can access Levoy's paper from the link.
- Other versions do not have the images from the original.
- To access click on Institutional Sign In



Transfer function (TF)

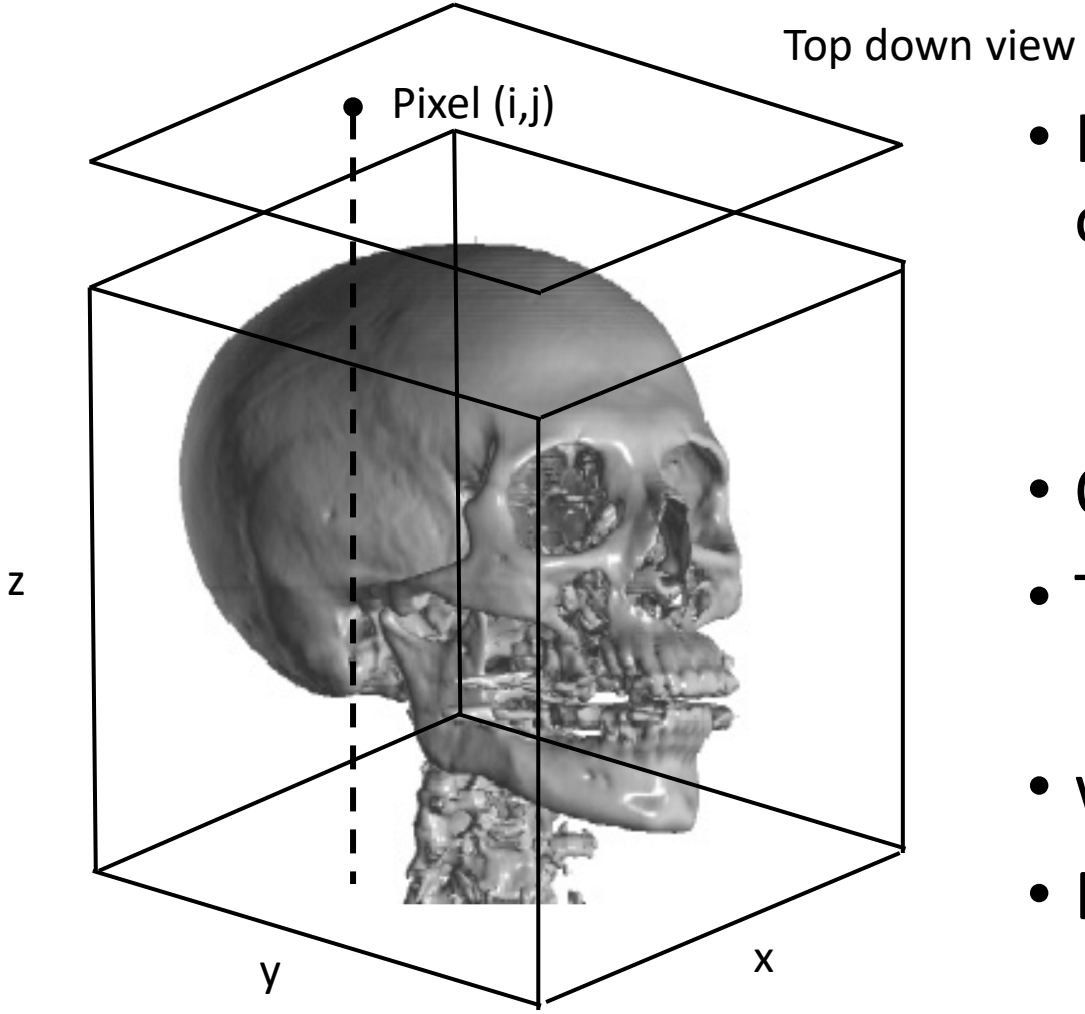
- The volume data contains an intensity value for each voxel.
- The values are e.g., X-ray absorption of the material.
- The transfer function maps voxel value to colour.
- We make a TF such that the colour matches the material (which had that voxel value).
- High quality images (like this one) require a good TF and good gradient shading.



Question: How do I start to code this?

- Let’s start in a way that we can develop the code in small steps.
- For each step we will get visual confirmation the algorithm is working so far (to make debugging easier than doing it all in one go).
- Step 1:
- Understand how to send rays through the volume.
- Step 2:
- While sending rays through the volume – test if there is a value >400 (bone) on our ray. If there is, make it white, otherwise, make it black.
- At this stage we should get a white head on a black background.

Casting rays through the volume



- Have the usual two outer loops to range over every pixel of the top down image

  for j=row 0 to image height

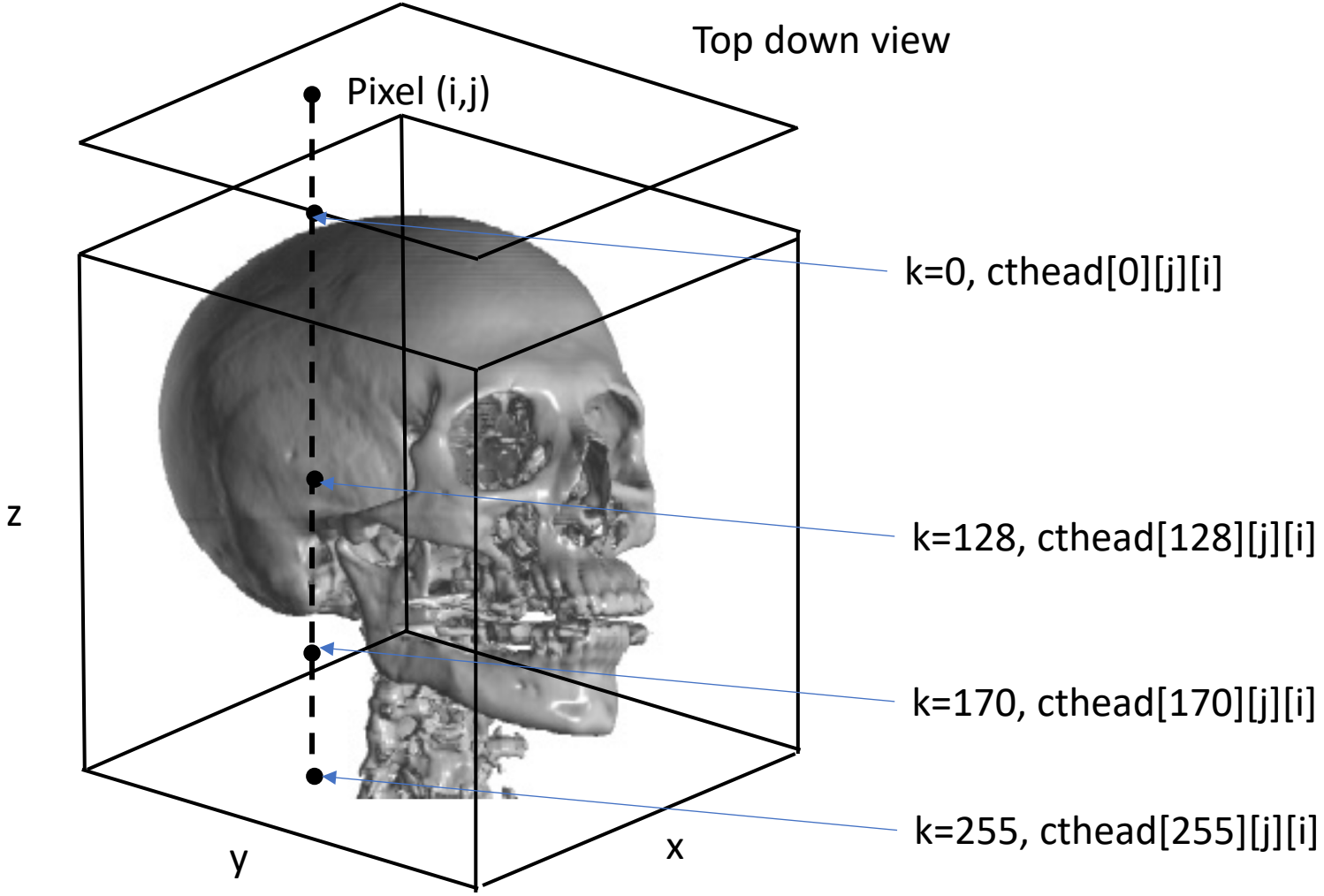
    for i=column 0 to image width

- One example pixel is shown (i,j)
- Then an inner loop

  for k=slice 0 to slice 255 (<=256)

- will range over every voxel along the ray
- Do a computation on cthead[k][j][i]

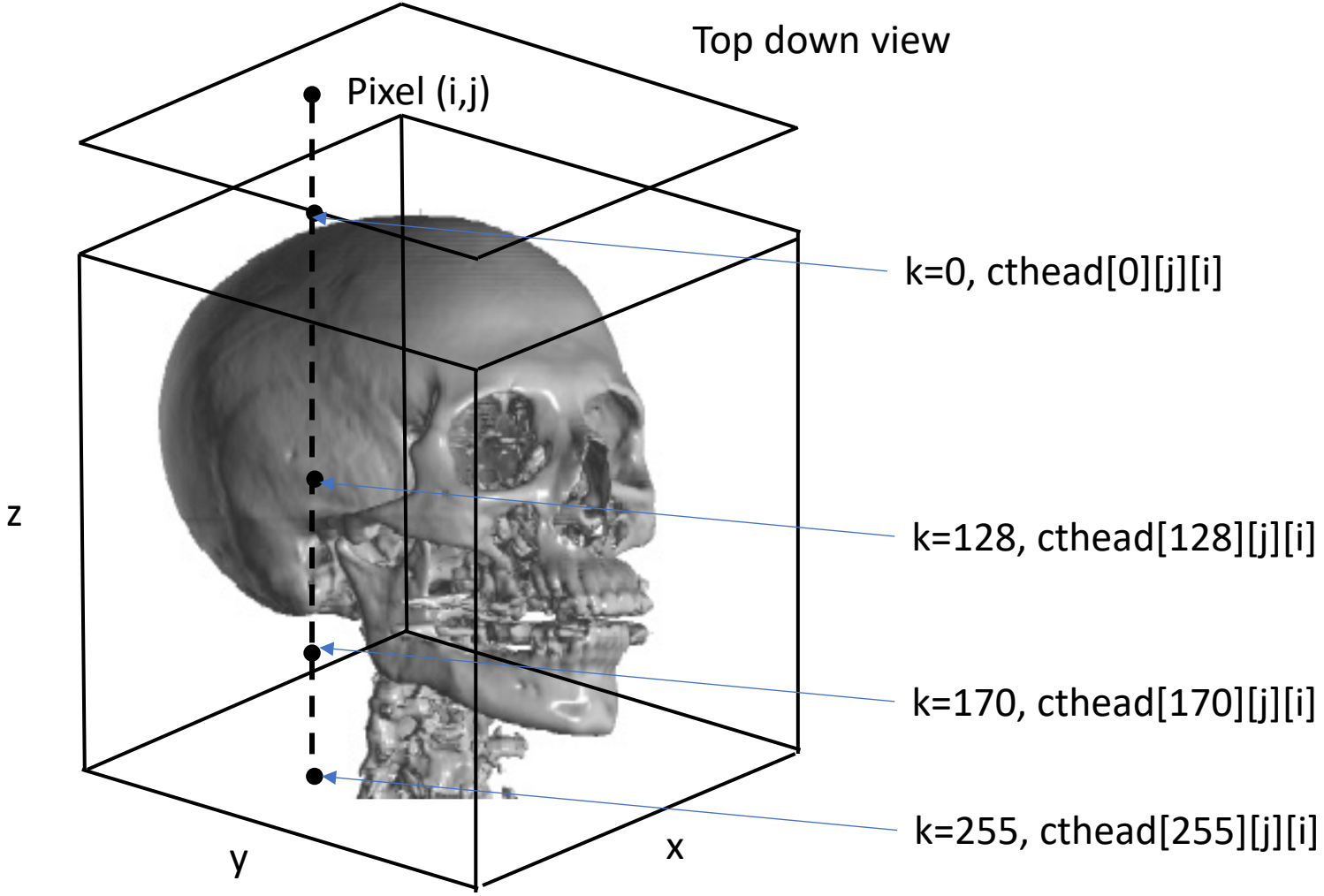
Casting rays through the volume



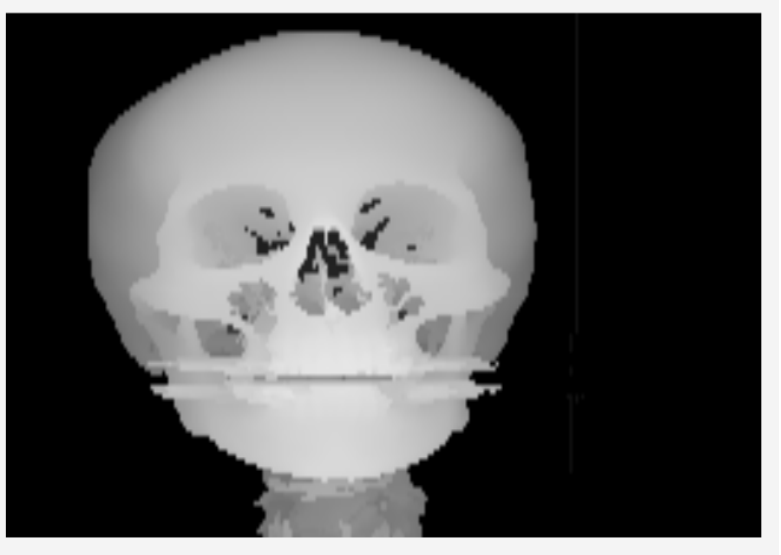
You can see the loop will visit each voxel along the ray.
Do a computation on cthead[k][j][i] to inform the colour at (i,j) on the image (top down view in this case).

Example, set a value to 0 before the loop and set it to 1.0 if cthead[k][j][i]>400 at any point in the loop.

Depth based shading

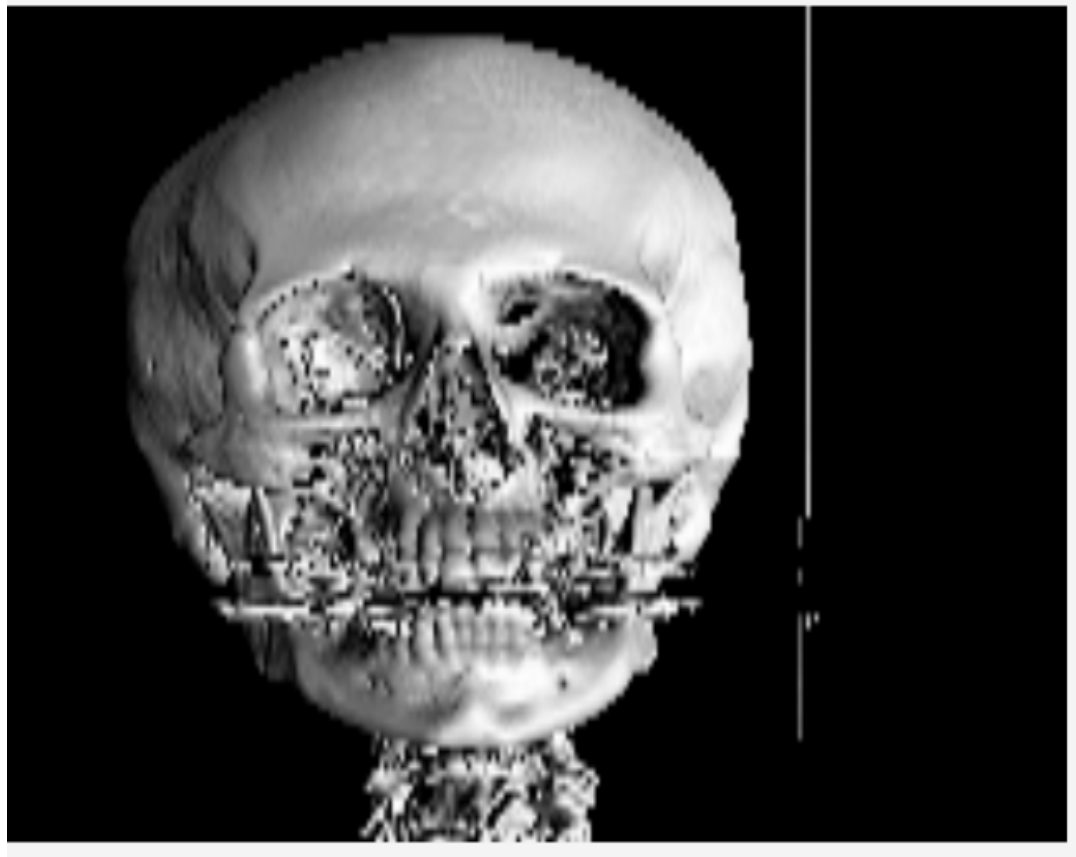


Depth based shading was developed early on in volume rendering.
The value of k for the top down view is 0 to 255 represents how far we are into the data.
Instead of setting the pixel to white, set it to k/255 for other views. Here's the front view with pixels set to 1.0-k/255.



Gradient shading

- At this stage we could extend it using gradient shading.
- Calculate the gradient. Normalise it. Take the dot product with the (normalised) vector to the light source. My light is to the middle right.
- This is not perfect at this stage because the hit points are the first integer position after the voxel goes above 400.
- We can find more accurate positions using interpolation.

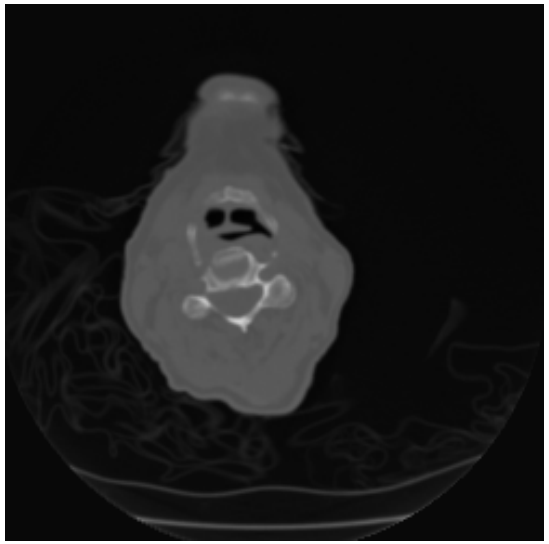
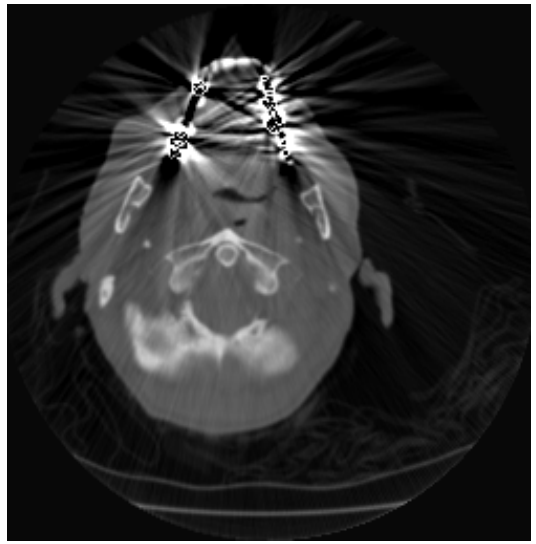
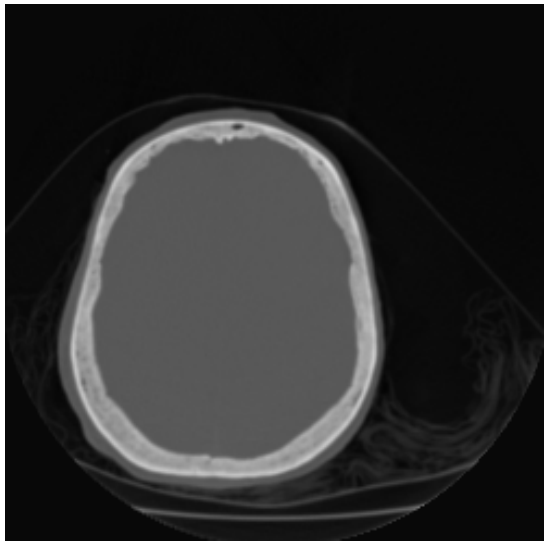


Gradient shading

- Find the 3D gradient G at sample locations (x,y,z)
- Use finite differences to obtain samples (see edge detection part of the course)
- E.g., we will use central differences:
- $G_x = getSample(x+1, y, z) - getSample(x-1, y, z)$
- $G_y = getSample(x, y+1, z) - getSample(x, y-1, z)$
- $G_z = getSample(x, y, z+1) - getSample(x, y, z-1)$
- Normalise the result
- Find the dot product between G and a vector to the light source for shading

Maximum Intensity Projection

- Type of rendering algorithm for volume data
- Typical slices from CThead

Maximum Intensity Projection

- Example for side view from CThead
- Rays are traced through the data set
- For each ray find the maximum value
- Then map it from [min,max] to [0,255]

Maximum Intensity Projection

- Algorithm (for 256x256x256 resampled data set – CThead):
  e.g. for top view, first loop over all the pixels:

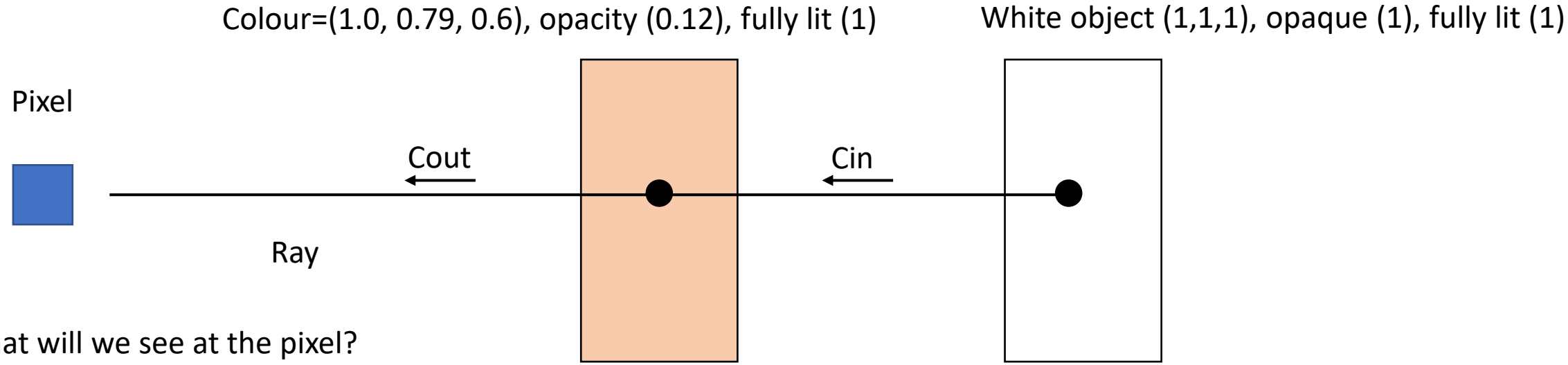
```
for j=row 0 to row 255
  for i=column 0 to column 255
    //now set the maximum to a default value
    maxSoFar=Integer.MIN_VALUE
    //now for the ray (i,j) going through the data in direction (0,0,1)
    //from slice 0 to slice 255, find the maximum
    for k=slice 0 to slice 255
          maxSoFar = max(cthead[k][j][i], maxSoFar)
    //normalise maxSoFar to give pixel colour
    //set the pixel colour and move onto next pixel
```

where max is a function that returns the maximum of its parameter list.

Maximum Intensity Projection

- Maximum intensity projection (MIP) has the advantage that it is close to a medical understanding of an X-Ray
- Its disadvantage is that the depth of the structure is hard to distinguish (sometimes depth weighting is used)

Compositing



What will we see at the pixel?
The first object is nearly transparent (low opacity). We will mostly see white opaque object.
Try to work out the equation.
We will find a recurrence relation. The pixel colour will be the light contributed by the first thing seen on the ray plus all the light from the objects behind (and so on).
Assume we already know the colour of everything behind the first thing to be seen and this is C^{in}. We want to find the combined colour (C^{out}) of the closest object combined with C^{in}. The opacity of the first object is σ. The lighting factor is L. L could be the depth we calculated in depth based shading, the diffuse illumination from gradient shading, or start off using L=1.0.

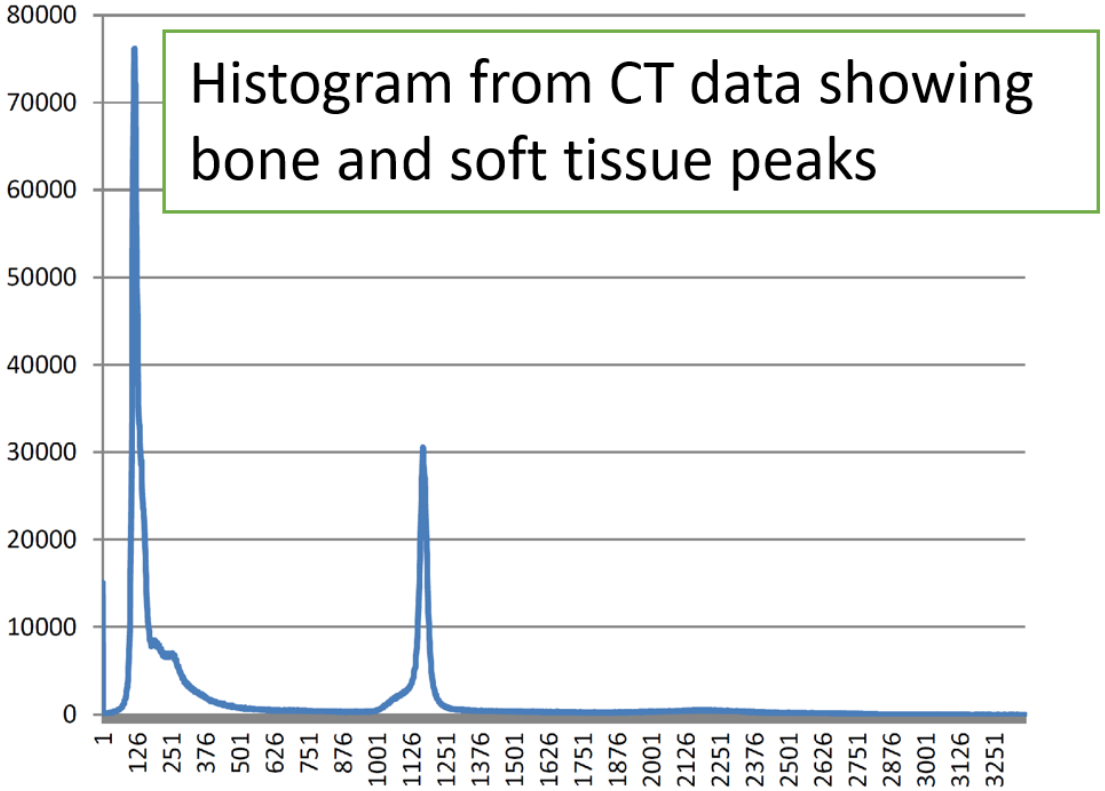
$$C^{out} = \sigma LC + (1 - \sigma)C^{in}$$

Transfer function (to find C and σ)

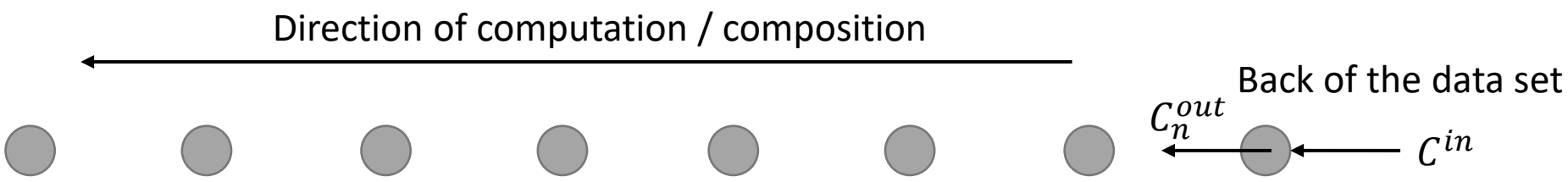
- cthead[k][j][i] gives us a value (Hounsfield value)
- We need a colour mapping from HU to colour and opacity
- The cthead data set has values -1117 to 2248 (resampled data set is -1897 3029)
- We need some code that sets the colour and opacity given a value in that range
- This code is up to you (e.g., you could use if..else statements in a function or a look up table (LUT)) – I prefer a LUT because it can be more efficient.
- E.g., for Cthead and the images shown later
- CT values below -300 were completely transparent (opacity=0).
- CT values from -300 to 49 were RGB=(1.0, 0.79, 0.6) and opacity = 0.12.
- CT values from 50 to 299 were completely transparent (opacity=0).
- CT values from 300 to 4096 were RGB=(1.0, 1.0, 1.0) and opacity = 0.8.

Histogram

- The histogram can provide information about the data



Back to front worked example



n^{th} Sample colour

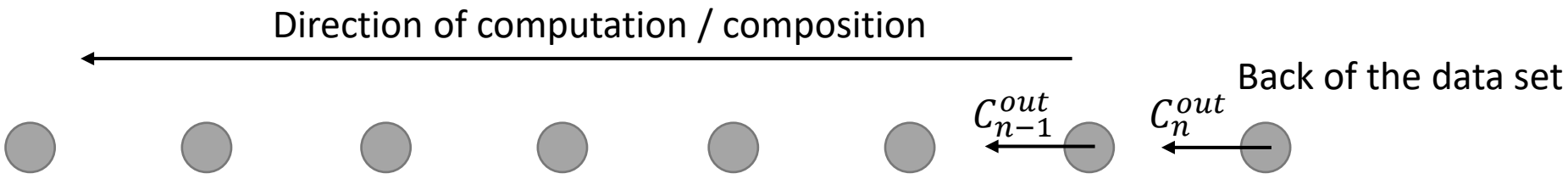
$$C_n = (r, g, b)$$

Sample opacity = σ_n

C^{in} into this sample is (0,0,0) because this is the "first" sample. I say "first" in quotes because it is the first one we encounter when looping from back to front – i.e. it is the last sample on the ray at the back, but the first we deal with in the back to front algorithm.

$$C_n^{out} = \sigma_n L C_n + (1 - \sigma_n) C^{in}$$

Back to front worked example



$n-1^{th}$ Sample colour

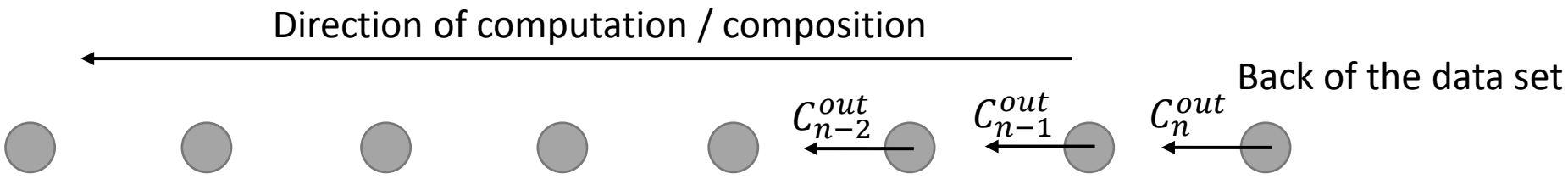
$$C_{n-1} = (r, g, b)$$

Sample opacity = σ_{n-1}

C^{in} into this sample the C_n^{out} from the previous sample.

$$C_{n-1}^{out} = \sigma_{n-1} L C_{n-1} + (1 - \sigma_{n-1}) C_n^{out}$$

Back to front worked example



$n-2^{th}$ Sample colour

$$C_{n-2} = (r, g, b)$$

Sample opacity = σ_{n-2}

C^{in} into this sample the C_{n-1}^{out} from the previous sample.

$$C_{n-2}^{out} = \sigma_{n-2} L C_{n-2} + (1-\sigma_{n-2}) C_{n-1}^{out}$$

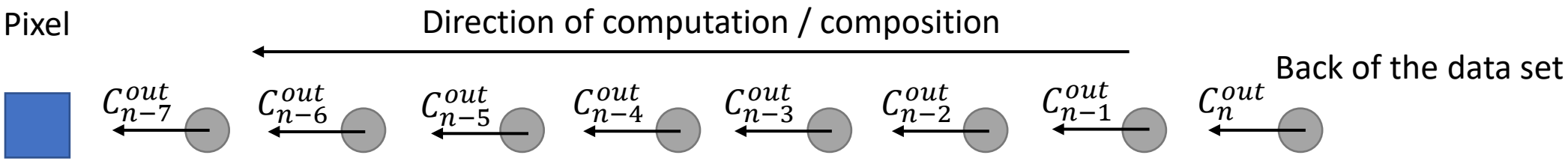
At each step we composite:

(1) the colour behind the current sample

(2) with the colour of the current sample

(3) in the ratio $\sigma_{n-2} : (1-\sigma_{n-2})$

Back to front worked example



$n-7^{th}$ Sample colour

$$C_{n-7} = (r, g, b)$$

Sample opacity = σ_{n-7}

C^{in} into this sample the C_{n-6}^{out} from the previous sample.

$$C_{n-7}^{out} = \sigma_{n-7} L C_{n-7} + (1 - \sigma_{n-7}) C_{n-6}^{out}$$

This is the compositing equation applied in back to front manner

$$C^{out} = \sigma L C + (1 - \sigma) C^{in}$$

Back to front implementation

Initialise accumulated colour: $Ca = (0, 0, 0)$

Loop over all the samples on the ray from back to front, doing:

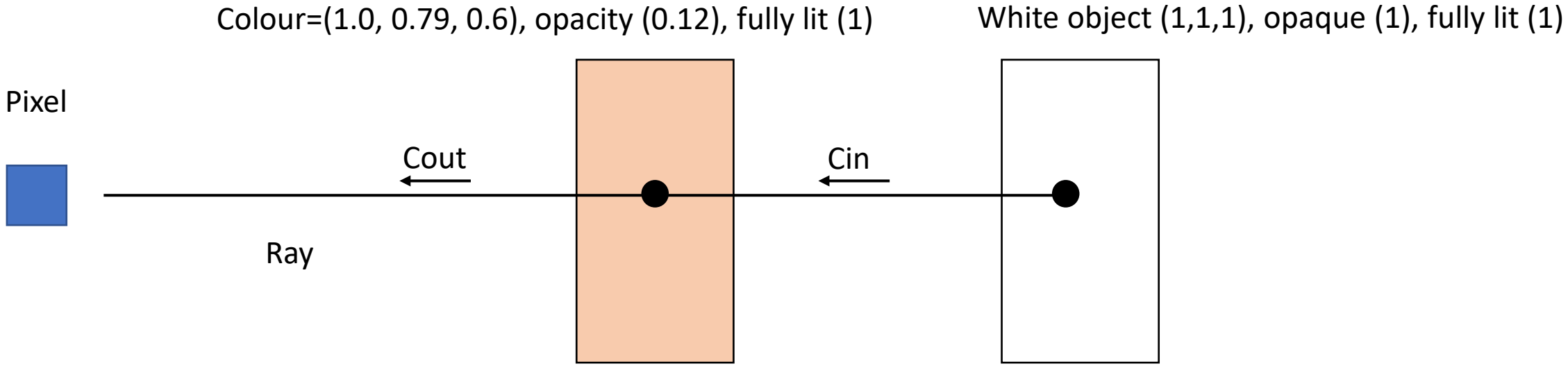
$Ca = \sigma LC + (1 - \sigma)Ca$

where L is the calculated light at the current voxel, C is the colour of the current voxel,

and σ is the opacity of the current voxel.

At the end of the loop, the final pixel colour is given by Ca.

Compositing (back to front = inefficient)



$$C^{out} = \sigma L C + (1 - \sigma) C^{in}$$

Imagine there are many samples behind the white object.
They will have no impact on the colour of the pixel because the white object is fully opaque.
So any compositing of those values is a waste of computation.
Back to front is easier to understand, but inefficient.

Compositing (front to back = efficient)

We will traverse samples from the pixel (front) to the back of the data.
We will accumulate the transparency of all the samples so far. (Note: transparency=1-opacity: $\alpha = (1 - \sigma)$).
At the start (before the ray casting loop), we initialise the transparency α^{accum}=1 (there is nothing in the way of the first sample).
We will accumulate the colour of all the samples so far.
At the start (before the ray casting loop), we initialise the accumulated colour C^{accum}=(0,0,0). No colour.
In the loop we will add α^{accum} of the contribution of the current sample σLC to the accumulated colour
Think about this:

For the first sample there is nothing in the way and we will see its full contribution.

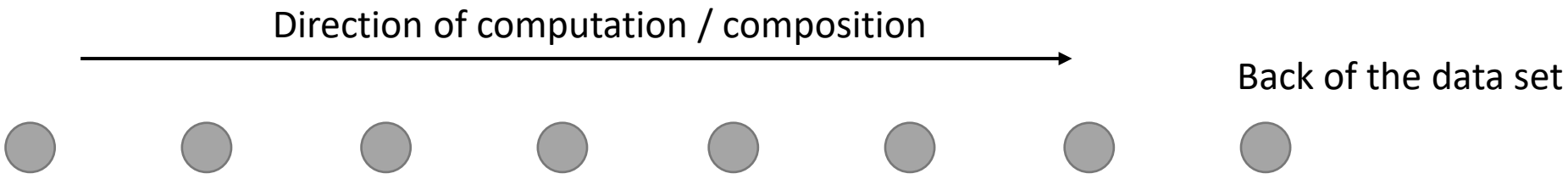
For the next sample we will see $(1 - \sigma)$ of its colour because the current sample has an opacity of σ.

So in the loop modify the accumulated transparency by how transparent the current sample is, $(1 - \sigma)$.
α^{accum}= $\alpha^{accum} \times (1 - \sigma)$
We can now continue the loop with the next sample

Initialise: α^{accum}=1, C^{accum}=(0,0,0)
Loop over all the samples on the ray, doing:
C^{accum}= $C^{accum} + \alpha^{accum}\sigma LC,\ \alpha^{accum} = \alpha^{accum} \times (1 - \sigma)$

Remember a=a+b can be written in Java, C and C++ as a+=b

Front to back

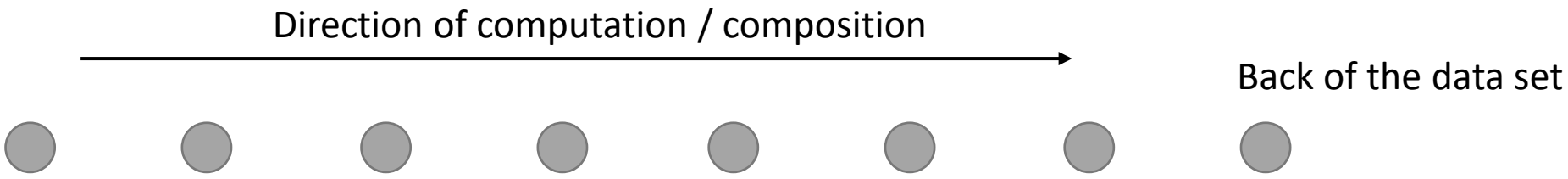


Initialise: α^{accum}=1, C^{accum}=(0,0,0)
L=1 in simple case (assume full lighting)

1st Sample colour: $C_1 = (r, g, b)$
Sample opacity = σ_1

$C^{accum} = C^{accum} + \alpha^{accum}\sigma_1 L C_1$,
$\alpha^{accum} = \alpha^{accum} \times (1 - \sigma_1)$

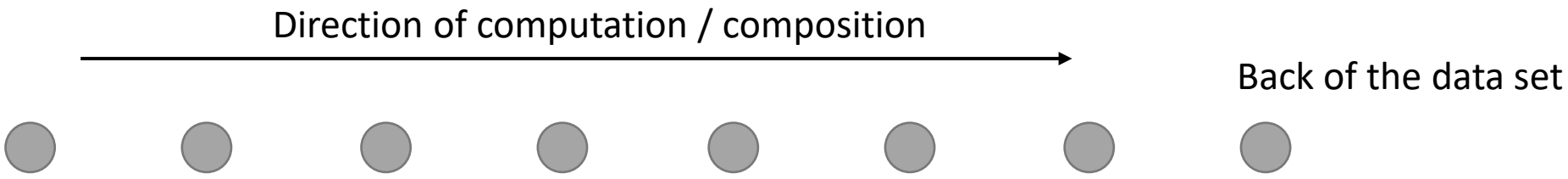
Front to back



2nd Sample colour: $C_2 = (r, g, b)$
Sample opacity = σ_2

$C^{accum} = C^{accum} + \alpha^{accum}\sigma_2 L C_2,$
$\alpha^{accum} = \alpha^{accum} \times (1 - \sigma_2)$

Front to back



3rd Sample colour: $C_3 = (r, g, b)$
Sample opacity = σ_3

$C^{accum} = C^{accum} + \alpha^{accum}\sigma_3 L C_3,$
$\alpha^{accum} = \alpha^{accum} \times (1 - \sigma_3)$

…and so on. The front to back approach uses the accumulated transparency to provide the equivalent result to back to front computation

Front to back implementation

Initialise: $a^{accum} = 1$, $C^{accum} = (0, 0, 0)$

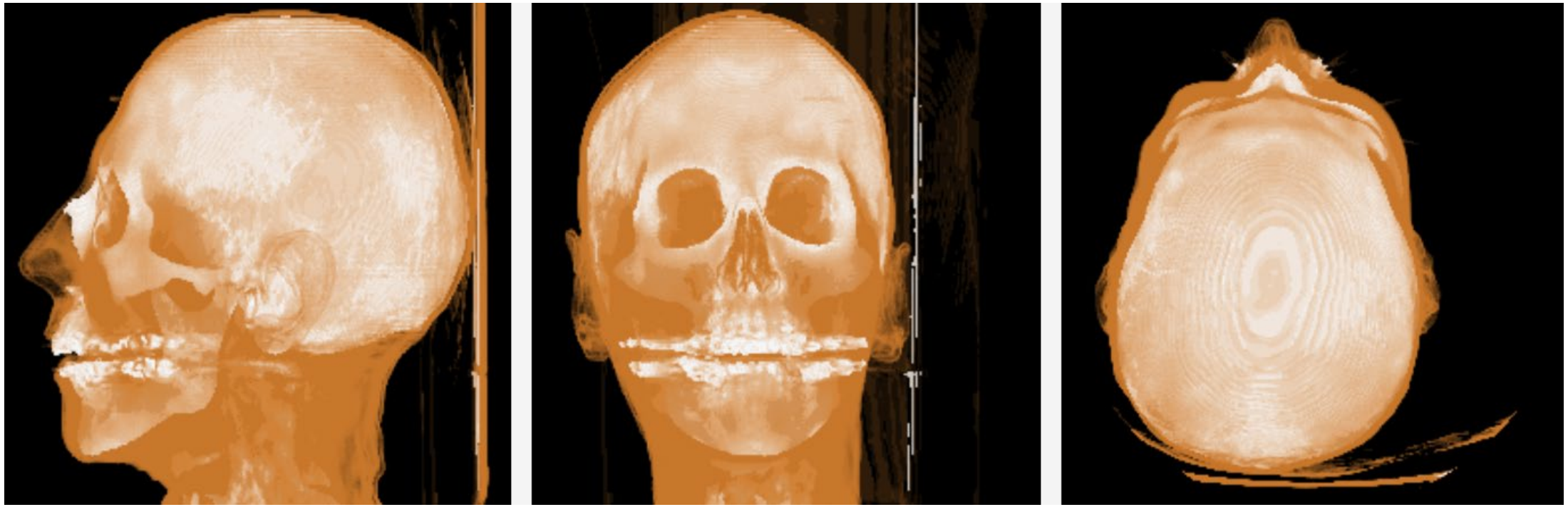
Loop over all the samples on the ray from front to back, doing:

$C^{accum} = C^{accum} + a^{accum}\ \sigma\ LC$

$a^{accum} = a^{accum} \times (1 - \sigma)$

- Early (or adaptive) ray termination (stopping the for loop) can happen when the accumulated transparency is zero (i.e., $a^{accum} = 0$) because at that point, no more colour will be added to the colour accumulator (C^{accum}).
- We can also (with a possible little loss of accuracy) terminate the ray when aaccum < τ where τ is some small user-defined threshold.
- This can be chosen to be small enough that it is likely there will be no discernable change of colour in the pixel (e.g., τ = 0.03).

- Top down, front-to-back
- This is what the images look like for front to back rendering. The top down view (left) has been done using front-to-back and the inner k loop has gone from zero to CT_z_axis.



Debugging graphics code

- You could print variable values to make sure they are as expected.
- If you put a print inside the inner ray loop, it will be printed millions of times (not helpful).
- Try, e.g.,

```
for (j=0; j<h; j++) {
   for (i=0; i<w; i++) {
      for (k=0; k<CT_z_axis; k++) {
         if (i==128 && j==128) {
            //print some values
```

Change pixel (128,128) to some other value of interest (i.e., this will not work on the shorter images).

Inside the inner k loop, using this approach, the values will be printed for every point on a single ray = CT_z_axis values (256)

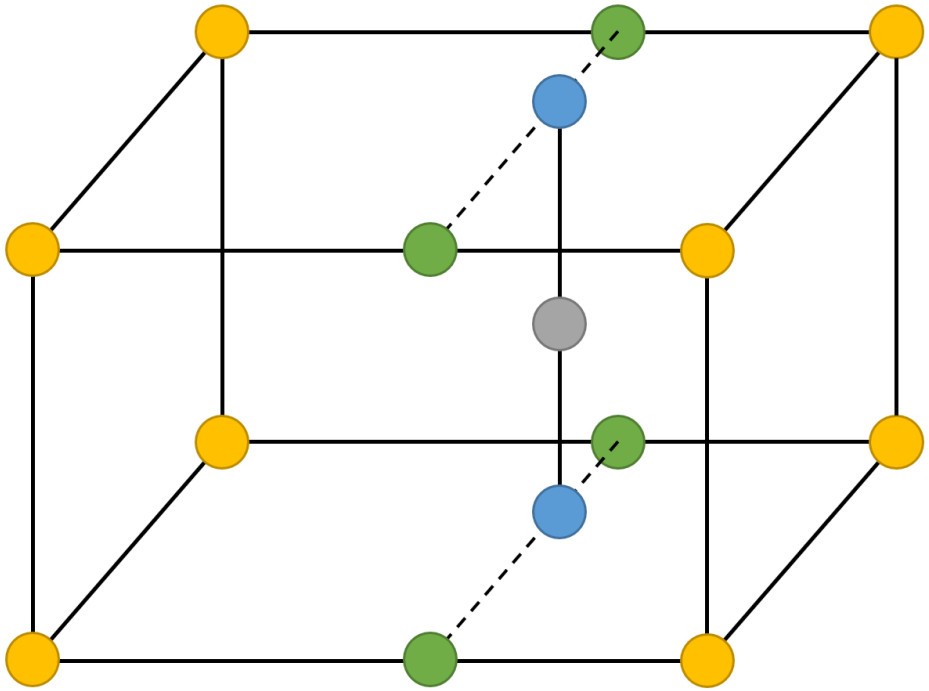
These will be more manageable to look through and debug.

Using a camera model (see ray tracing)

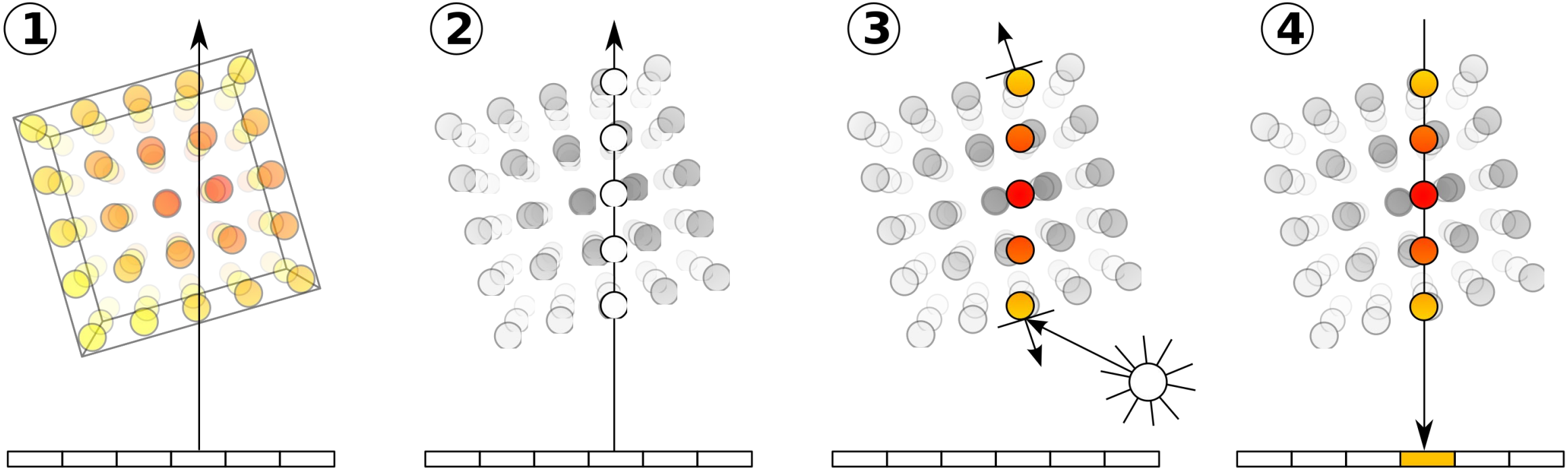
```
Set up the camera (VRP, VPN, VUV, VRV)
For each pixel (i,j)
      Calculate (u,v) from (i,j)
      Calculate ray_origin from (u,v) and camera parameters
      Initialise C_a=(0,0,0)
      Find Ray-AABB intersections (t1 and t2)
      For t=t2 downto t1
            v=getSample(x, y, z)
            Obtain voxel colour C and opacity o from the sample value v
            Calculate the surface normal at t, and hence the value for L
            C_a=o*C*L+(1-o)*C_a
      Pixel colour is set to C_a
```

Trilinear interpolation

For floating point real sample locations use trilinear interpolation
Gold voxel values known
Interpolate green ones along x-axis
Interpolate blue ones from green ones along y-axis
Interpolate grey one from blue ones along z-axis



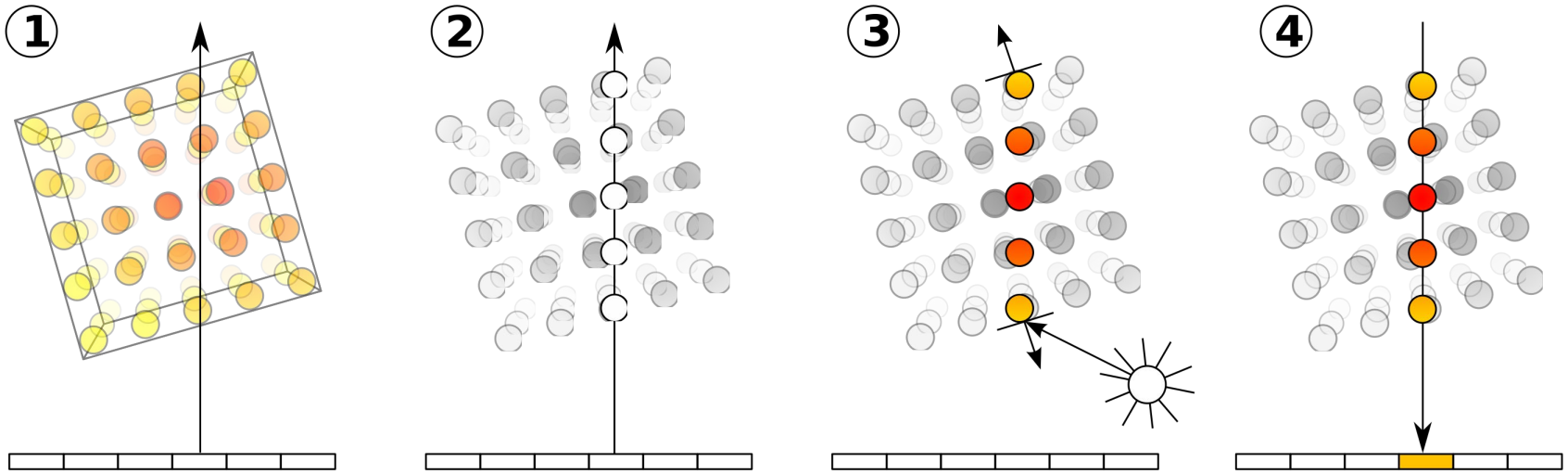
Volume Rendering (volume ray casting)



Volume_ray_casting.png: Thetawavederivative work: Florian Hofmann - Volume_ray_casting.png, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=14521474

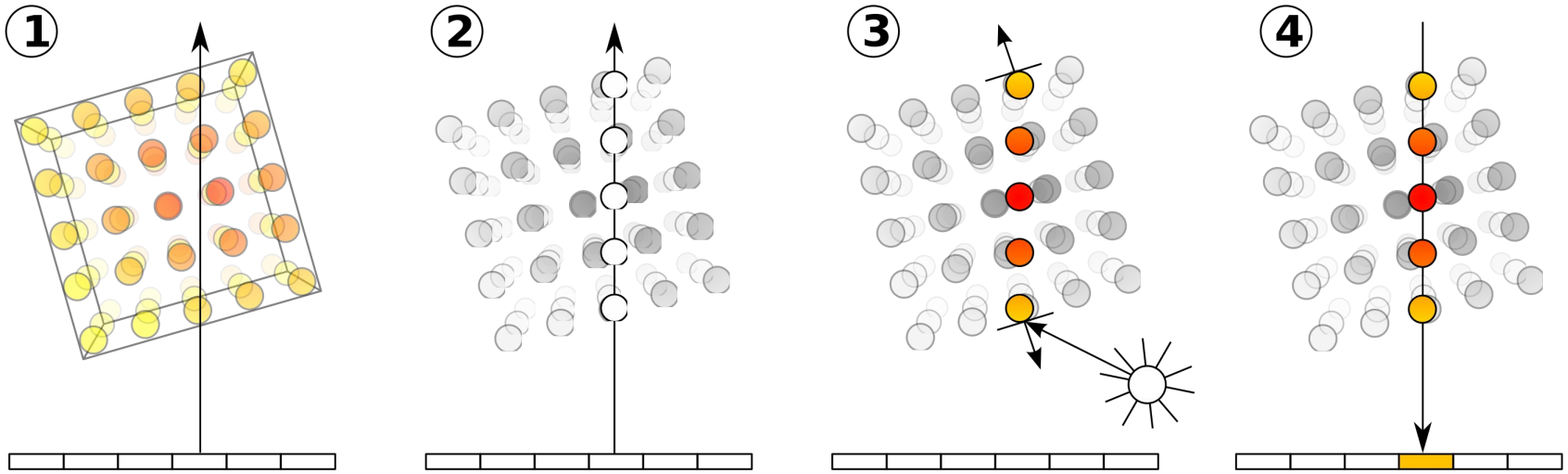
Four steps (adapted from Wikipedia)

1. **Ray casting**. For each pixel of the final image, a ray of sight is shot ("cast") through the volume. At this stage it is useful to consider the volume being enclosed within an axis-aligned bounding box (AABB) and use a Ray-AABB intersection test. *For the assignment we will not need to intersect with the volume because our cameras are fixed to top-down, side or front views and thus we can avoid this.*



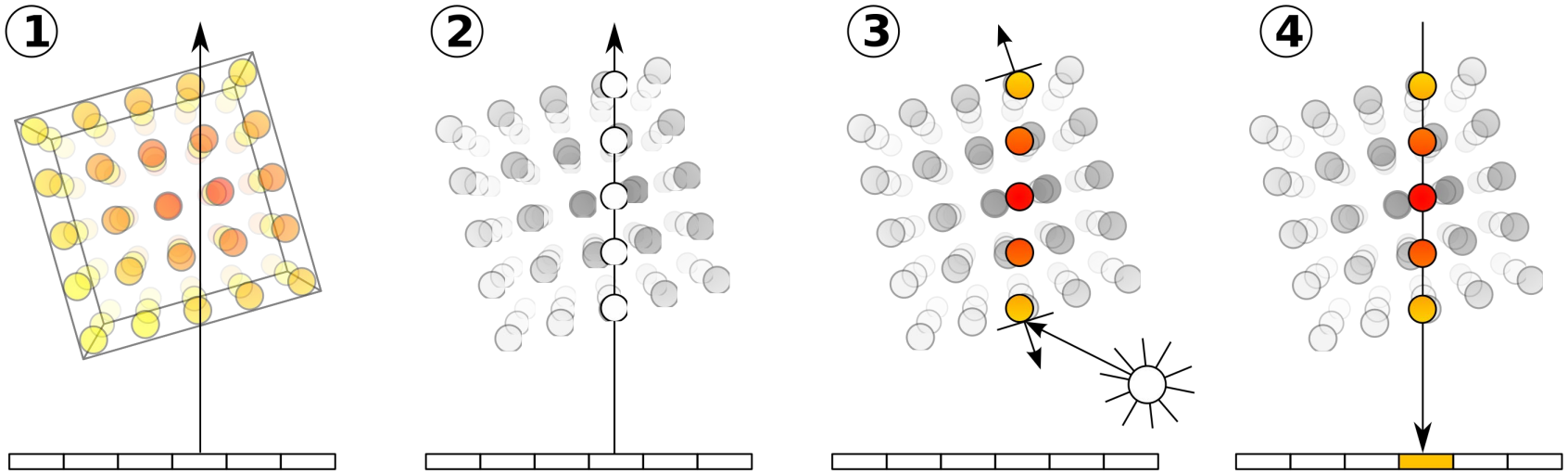
Four steps (adapted from Wikipedia)

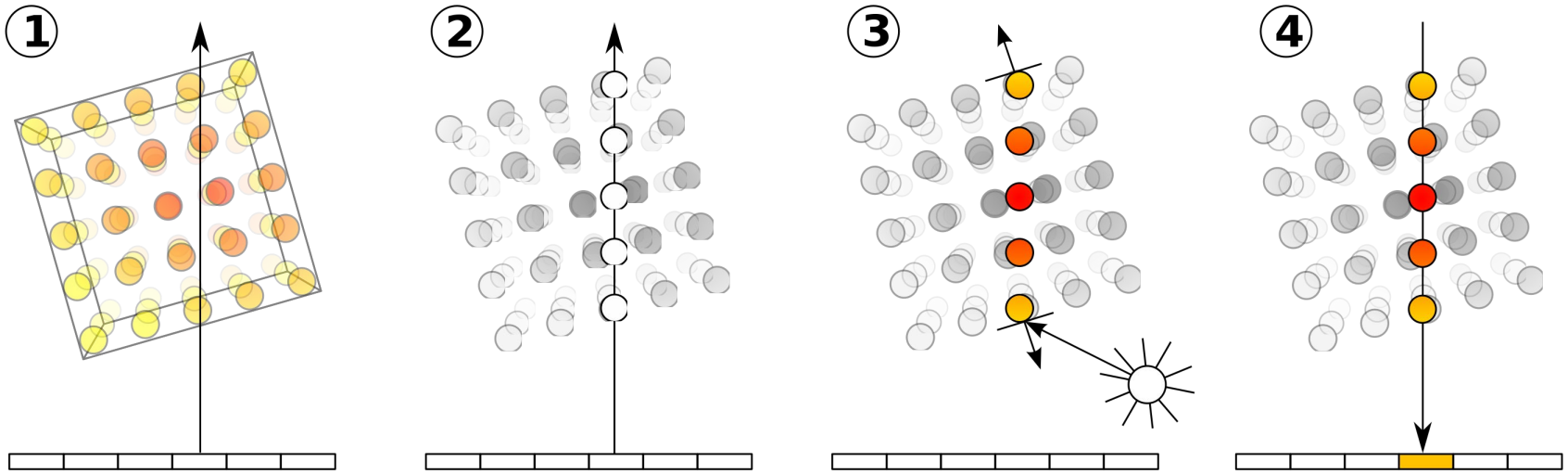
2. **Sampling**. Along the part of the ray lying within the volume, equidistant samples are selected. *In the assignment, these are integer positions e.g., Cthead[z][y][x] where (x,y) are the pixel being traced in the top-down view and z is the current slice number being evaluated*. *We do not need to deal with the general case:* In general, the volume is not aligned with the ray of sight, and sampling points will usually be located in between voxels. Because of that, it is necessary to interpolate the values of the samples from its surrounding voxels (commonly using trilinear interpolation). *Interpolation is covered in the "Interpolation" lectures and chapter.*



Four steps (adapted from Wikipedia)

3. **Shading**. For each sampling point, a transfer function retrieves an RGBA material colour and a gradient is computed. The gradient represents the orientation of local surfaces within the volume. The samples are then shaded (i.e. coloured and lit) according to their surface orientation and the location of the light source in the scene.



Four steps (adapted from Wikipedia)

4. **Compositing**. Colours for each sample are composited. Think of each sample as a piece of stained glass with lots of them in a row. It may work back-to-front, i.e. farthest sample first (easier to understand). Or, more efficiently, front to back.